Module 10. Foodborne Diseases

Lesson 10.1:

Investigation of Foodborne Illness

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Learning Outcomes

1. Define terminologies related to foodborne diseases
2. Identify foodborne pathogens and chemical agents that cause food infection and intoxication.
3. Discuss the important elements in conducting foodborne illness investigation.
4. Identify risk factors predisposing to foodborne illness.
5. Describe general and specific signs of foodborne illnesses

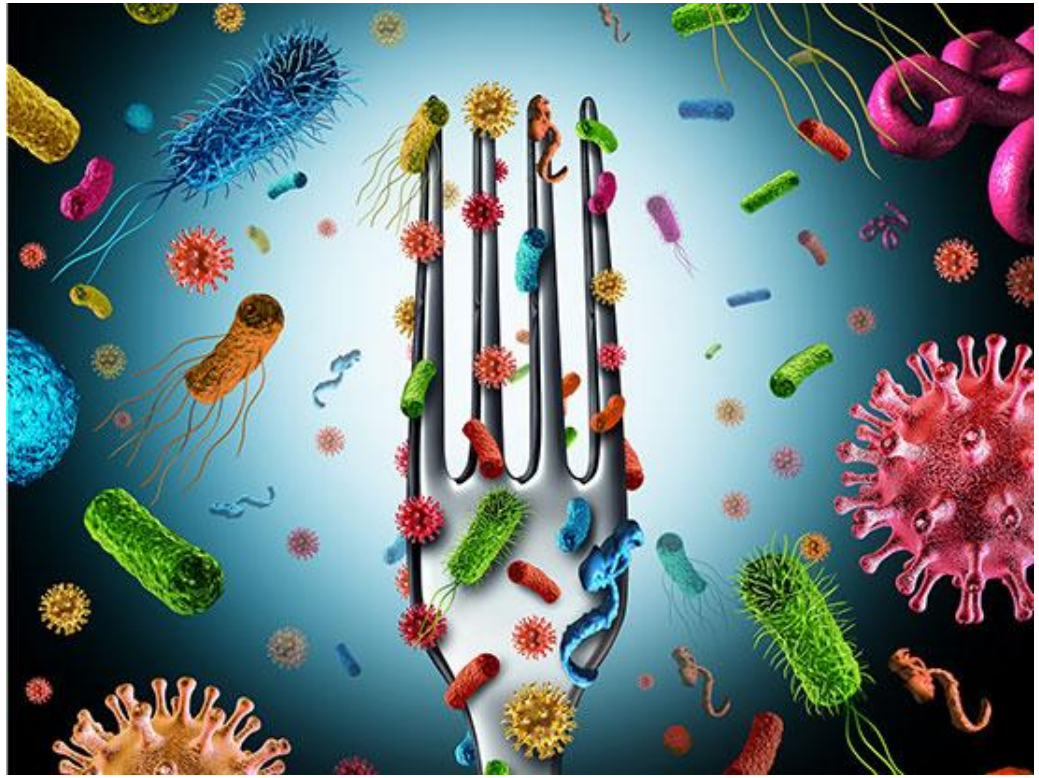
Foodborne Illness

- illness/es acquired through consumption of food and water contaminated with pathogens or certain chemicals
- caused by bacteria, viruses, molds, worms, and protozoa
- chemicals:
  - natural components of foods
  - accidentally added during production and processing
  - carelessness or pollution

- The main causes of foodborne illness are:
  - bacterial (66%)
  - chemical (26%)
  - viral (4%)
  - parasitic (4%)

- Food allergy is an abnormal response to a food triggered by the body's immune system.

- Allergens: nuts, milk, eggs, fish, crustacean shellfish, tree nuts, peanuts, wheat or soybeans, can cause allergic reactions in people with food allergies.



https://www.uab.edu/news/youcanuse/item/8792-what-s-making-you-sick-an-in-depth-look-at-food-borne-illnesses



https://www.nestlenutrition-institute.org/

Classification of foodborne illness

- **Foodborne infection**
  - caused by the ingestion of food contaminated with viable pathogenic organisms that can establish infection
  - after ingestion, the organism colonizes the digestive tract and multiplies in number, then spread to other parts of the body via blood circulation
- **Foodborne intoxication/poisoning**
  - occurs when pre-formed toxins produced by pathogens are ingested
  - may include intoxications from poisonous plants or toxic animal tissues, or due to consumption of food contaminated by chemical poisons
- **Toxin-mediated infection**
  - caused by toxin-producing organism but the toxin is produced inside the body (eg. *Cl. perfringens*)

- Food poisoning outbreaks are usually recognized by the occurrence of illness within a short period of time among individuals who have consumed one or more foods in common
- Diagnosis is suggested by epidemiological findings
- Single cases of food poisonings are difficult to identify unless, as in botulism, there is a distinctive clinical syndrome



Food infection

- *Campylobacter jejuni*
- **Escherichia coli** 0157:H7
- Hepatitis A virus
- *Listeria monocytogenes*
- Norwalk virus
- *Salmonella*
- *Shigella*
- *Vibrio parahaemolyticus* and other marine *Vibrio* spp.

Food intoxication

- *Bacillus cereus*
- *Clostridium botulinum*
- *Clostridium perfringens* (cafeteria germ)
- *Staphylococcus aureus*
- toxigenic molds (mycotoxins), poisonous mushrooms, and biogenic amines

Investigation of Foodborne Illness

Aims of an investigation of foodborne illness

1. identify offending foods and beverages
2. elucidate information about causative agents and their source
3. determine the factors that contribute to growth and survival of causative agents



- The investigation involves
  - **Patient**
  - **Food**
  - **Place** where the food is processed, prepared and served







Patient



- record the clinical symptoms of the affected persons
- determine the incubation period
- obtain clinical specimens as soon as possible such as vomitus, stool and/or blood according to signs and symptoms

Food

- collect information about all meals and snacks eaten 72 hour before onset of illness
- collect leftovers from food that were eaten 72 hour before onset of illness or some ingredients that were used in foods and of any food available from an allegedly contaminated lot





Place

1. Investigate the hygienic condition at the place where the suspect foods were processed, prepared and served.

   Try to identify the circumstances that permitted survival and growth of food-borne pathogens such as:

   i) inadequate refrigeration

   ii) inadequate cooking and reheating

   iii) holding foods at warm temperature several hours before serving

   iv) inadequate cleaning and disinfection of equipment

   v) the possibility of cross contamination from raw to cooked foods

2. Arrange for clinical and laboratory examination of workers who ate the suspect food.
3. Submit a report covering all information obtained.

Summary of steps in the investigation of food-borne disease outbreak investigation

1. Verify the existence of an outbreak
   - Compare the current number of cases with the past

     Note: -consider seasonal variations

2. Verify the diagnosis
   - Review clinical and laboratory findings
3. Describe the outbreak with respect to time, place and person
4. Prepare an epidemic curve
5. Calculate food-specific attack rates
6. Formulate and test hypotheses (by consulting with higher level health professionals).

7. Search for additional cases

8. Analyze the data

9. Make a decision on the hypotheses tested (by consulting with higher level health professionals).

10. Intervene and follow-up

11. Report the investigation

12. Inform the public on the control and prevention of the outbreak

Top 5 Foodborne Illness Risk Factors

1. **Poor Personal Hygiene**

- Poor personal hygiene practices serve as the leading cause of foodborne illnesses. Food establishments must promote a culture of food safety by developing an employee illness policy, proper handwashing procedure, and a no barehand contact policy with ready to eat foods

2. **Improper Holding Temperatures**

- Cold foods shall maintain ≤41°F and hot foods shall maintain ≥135°F. If foods stay in between those temperatures for any period, ensure they are time and temperature controlled and documented. Time and temperature control for safety (TCS) foods must be date marked and stored no longer than 7 days. Remember, first day of preparation or when a food package is opened, counts as day 1. When in doubt, throw it out!

3. **Improper Cooking Temperatures**

- Ensure metal stem thermometers are conveniently stored and accessible for employees to monitor final cooking temperatures.

4. **Food from Unsafe Sources**

- All foods distributed in licensed food establishments must be obtained from approved sources that comply with applicable laws and regulations. All food shall be inspected upon delivery to ensure proper temperatures, it’s condition and overall sanitation.

5. **Contaminated Equipment/Cross-Contamination**

- Cross contamination can occur from a variety of sources, including chemical and raw foods.
- To prevent the spread of harmful pathogens
  - all equipment and utensils shall be properly cleaned and sanitized at least once every 4 hours
  - ensure sanitizer solutions are monitored using their appropriate test strips
  - low sanitizer concentrations will fail in effectively removing pathogens from surfaces, while high concentrations will leave a toxic residue

Effects of food poisoning

- develop symptoms after ingesting a pathogen
- others never know that they are suffering from food-borne illness
- High risk:
  1. compromised immune systems
  2. elderly
  3. very young



- Symptoms of foodborne illnesses depend on the cause
- Common symptoms of many foodborne illnesses include **vomiting, diarrhea or bloody diarrhea, abdominal pain, fever, and chills**
- Foodborne illnesses may lead to dehydration, hemolytic uremic syndrome (HUS), and other complications
- Acute foodborne illnesses may also lead to chronic— or long lasting—health problems

- Treatment: replacing lost fluids and electrolytes to prevent dehydration.
- Foodborne illnesses can be prevented by properly storing, cooking, cleaning, and handling foods.

Lesson 10.2:

Bacterial, Viral, Parasitic, and Chemical Foodborne Diseases

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Learning Outcomes

1. Discuss etiologic agent, transmission, implicated foods and means of control for specific foodborne diseases.
2. Tabulate foodborne diseases based on classification.

Why do foodborne diseases emerge?

- increase in international travel and trade
- microbial adaptation
- changes in the food production system
- changes in human demographics and behavior

Other contributing factors

- **The globalization of the food supply**
  - Importation of contaminated raspberries from South America led to a large outbreak of cyclosporiasis in North America in 1996-97
- **Inadvertent introduction of pathogens into new geographic areas**
  - Introduction of Vibrio cholerae into waters off the coast of southern United States when a cargo ship discharged contaminated ballast water in 1991
- **Travelers, refugees, and immigrants exposed to unfamiliar foodborne hazards while abroad**
  - International travelers may become infected by foodborne pathogens that are uncommon in their countries
  - 90% of all cases of salmonellosis in Sweden are imported

- **Changes in microorganisms**
  - evolution of new pathogens
  - development of new virulent strains in old pathogens
  - development of antibiotic resistance pathogens
  - ability to survive in adverse environmental conditions
- **Change in the human population**
  - population of highly susceptible persons is expanding world-wide
    - ageing, malnutrition, HIV infections and other underlying medical conditions
    - reduced immunity due to poor nutritional status render people more susceptible to foodborne infections

- **Changes in lifestyle**
  - Greater numbers of people go out and eat meals prepared in restaurants, canteens, fast food outlets, and by street food vendors
  - Boom in food service establishments is not matched by effective food safety education and control
  - Unhygienic preparation of food provides ample opportunities for contamination, growth, or survival of foodborne pathogens

DISEASE	PATHOGEN	INCUBATION PERIOD/MEAN	DURATION OF ILLNESS	SYMPTOMS	FOODS IMPLICATED	TYPE OF ILLNESS	RESERVOIR
Salmonellosis	*Salmonella* (facultative) BACTERIA	8-72 hours 18-36 hours	2-3 days	Abdominal pain, headache, fever, nausea, diarrhea, chills, cramps	Poultry, eggs, meat, meat products, milk, smoked fish, protein foods, juice	Infection	Domestic and wild animals, humans
Shigellosis	*Shigella* (facultative) BACTERIA	1-7 days 1-3 days	Indefinite	Diarrhea, fever, chills, cramps, lassitude, nausea, dehydration	Moist, mixed foods, milk, potato, tuna, bean, and macaroni salads, apple cider	Infection	Human feces, flies
Listeriosis	*Listeria monocytogenes* BACTERIA	2 days – 3 weeks 4-21 days	Indefinite; high fatality in the immunocompromised	Nausea, vomiting, chills, headache, fever, backache, meningitis	Unpasteurized dairy foods, vegetables, meat, poultry, seafood, raw and smoked fish, ready-to-eat foods	Infection	Domestic and wild animals, humans, soil
Staphylococcus	*Staphylococcus aureus* BACTERIA	1-6 hours 2-4 hours	1-2 days	Nausea, cramps, vomiting, dehydration	Reheated foods, ham, poultry, dairy foods, custards, potato salad, cream-filled foods	Intoxication	Humans (skin, throat, nose), animals
Clostridium perfringens	*Clostridium perfringens* BACTERIA	8-22 hours 10 hours	24 hours	Abdominal pain, diarrhea, gas	Meat, meat products, gravy, thick foods that are held at a low temp or cooled slowly	Infection/ Intoxication	Human intestinal tract, animals, soil
Bacillus cereus	*Bacillus cereus* BACTERIA	½ - 5 hours; 8-16 hours ½ -5 hrs; 12 hrs	6-24 hours; 12 hours	Nausea and vomiting; Diarrhea, abdominal cramps	Rice, custards, seasonings, dry food mixes, spices, salads, casseroles, milk, sauces,	Intoxication	Soil, dust

					meats		
Campylobacter	*Campylobacter jejuni* BACTERIA	2-5 days	5-7 days 3-5 days	Diarrhea which may be watery or bloody, fever, nausea, headache	Chicken, unpasteurized milk, water	Infection	Domestic and wild animals (intestinal tract)
E. coli 0157:H7	*Escherichia coli* BACTERIA	2-5 days	5-10 days 2-5 days	Diarrhea (may be bloody), abdominal pain, nausea, vomiting, fever, HUS	Undercooked ground beef, imported cheeses, unpasteurized milk or juice, cider, alfalfa sprouts	Infection/ Intoxication	Human intestinal tract, animals (especially cattle)
Botulism	*Clostridium botulinum* BACTERIA	12-36 hours; 72 or more hours	Several days – year 18-36 hours	Vertigo, inability to swallow, respiratory paralysis, visual disturbances	Improperly processed low-acid canned goods, garlic in oil, grilled onions, meat, stew, baked potatoes	Intoxication	Soil, water
Viral Gastroenteritis	Snow Mountain, calicivirus virus	24-48 hours	24-60 hours 36 hours	Diarrhea, vomiting, nausea, cramps, headache, muscle aches, tiredness, slight fever	Water, ready-to-eat foods (salad, sandwiches, bread) shellfish	N/A	Human intestinal tract
Hepatitis A	Hepatitis A VIRUS	7-50 days	Several weeks to months 25-30 days	Fever, fatigue, stomach pain, nausea, dark urine, jaundice	Ready-to-eat foods, fruit and juice, milk products, shellfish, salads, vegetables, sandwiches, water	N/A	Infected people
Norovirusn infection	Noro Virus	12-96 hours	1-2 days	Low-grade fever, chills, headache, muscle ache, and fatigue	Ready-to-eat foods (salad, sandwiches, bread) shellfish	Infection	Human intestinal tract

Also read PDF copy of:

Foodborne Illness Quick Reference

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Page 3	Bacillus cereus
Page 4	Botulism
Page 5	Campylobacteriosis
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